

Notes: Abrardi, Colombo, and Tedeschi (RFS, 2025)

Pretending Ignorance Is Bliss: Competing Insurers with Heterogeneous Informational Advantages

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1 Big picture

- **Question.** What happens in a competitive insurance market when insurers, rather than policyholders, hold private information about policyholder risk, and when insurers' signals are imperfect and heterogeneous?
- **Modern motivation.** Big data, machine learning, claims histories, and proprietary underwriting models make insurers increasingly better informed than policyholders about risk. But insurers do not necessarily have identical data, algorithms, or predictive accuracy, so their risk assessments can differ.
- **Core model.** Two risk-neutral insurers receive independent private signals about whether the environment is safe or dangerous. A risk-averse policyholder does not observe these signals. Insurers compete by offering exclusive, nonwithdrawable insurance contracts.
- **Key distinction.** Equilibria can be *noninformative*, where insurers pool and their offers reveal nothing about their signals, or *informative*, where menus separate insurer types and reveal market information to the policyholder.
- **Main noninformative result.** Noninformative equilibria can be strictly profitable, can involve risk pooling, and can even be ex ante efficient. This differs from classic Rothschild-Stiglitz adverse-selection logic, where pooling is usually vulnerable to cream-skimming deviations.
- **Main informative result.** Informative equilibria involve a novel interaction between signaling and screening. Once insurers reveal information through their menus, the policyholder becomes more informed than insurers about the market signal profile at the time of contract choice. This endogenous screening problem generates underinsurance and inefficiency.
- **Why profits survive competition.** Competition is weakened because deviations are interpreted as signals. If the policyholder has optimistic out-of-equilibrium beliefs, undercutting deviations may be rejected. In informative equilibria, incentive compatibility constraints also prevent some competitive deviations.
- **Signal precision result.** Higher signal precision changes premiums and profits in nontrivial ways. In noninformative equilibria, it benefits insurers with favorable risk estimates and hurts those with unfavorable estimates. In informative equilibria, the sign reverses because precision also changes the probability of signal profiles.
- **Extensions.** The paper studies withdrawable contracts, more than two insurers, and two-sided asymmetric information. The qualitative message survives: information revelation is not automatic, and competition may sustain profitable pooling.
- **Economic takeaway.** In markets where firms are better informed than consumers, competition need not force full information revelation or zero profits. Sometimes insurers can profitably “pretend ignorance is bliss” by pooling and not revealing what they know.

2 Related literature and contribution

2.1 Classic insurance models

- The classic Rothschild and Stiglitz (1976) model studies adverse selection when policyholders know their risk and insurers do not.
- In that environment, competitive insurers offer contracts to screen policyholder types. Pooling contracts are vulnerable to cream-skimming deviations that attract only low-risk policyholders.
- Abrardi, Colombo, and Tedeschi reverse and enrich the information structure: insurers have private, imperfect, heterogeneous information about risk.

Interpretation: The paper is not simply a variant of Rothschild-Stiglitz. The informed side of the market is different, and this changes what deviations communicate.

2.2 Informed-principal and signaling insurance models

- The paper builds on informed-principal models in which contract offers signal information held by principals.
- Villeneuve (2005) studies competitive insurers with identical perfect information about policyholder risk.
- Abrardi, Colombo, and Tedeschi add dispersed and imperfect signals, so insurers may disagree about the risk environment.

Interpretation: Heterogeneous insurer information is important because it creates signal profiles. The policyholder may learn more from the market configuration of offers than any single insurer knew when designing its menu.

2.3 Why the contribution matters

- The model can rationalize profitable insurance contracts in competitive markets.
- It can rationalize why insurers sometimes do not use or reveal all observable risk-related information in pricing.
- It links data analytics and insurance market outcomes: better insurer information may raise premiums, change profits, and affect whether information is revealed.
- It provides a new mechanism for market concentration: when too many insurers compete, equilibria may fail to exist because deviation constraints become too stringent.

Interpretation: The central novelty is the interaction of signaling by insurers and screening by policyholders, produced endogenously by market information.

3 Model primitives and notation

3.1 Agents and states

- There are two insurers, $i \in \{1, 2\}$.
- The policyholder has initial wealth W and faces a loss $L \in [0, W]$.
- There are two payoff states: loss ℓ and no loss $n\ell$.
- The probability of loss depends on the environment $\theta \in \{d, s\}$, where d denotes dangerous and s denotes safe.
- Loss probabilities satisfy $p_d > p_s$.
- The dangerous environment occurs with probability q ; the safe environment occurs with probability $1 - q$.

Interpretation: The environment is a common-value component of risk. It affects both insurers' expected profits and the policyholder's value of insurance.

3.2 Contracts

An insurance contract is

$$c = (w_\ell, w_{n\ell}) \in \Omega, \tag{1}$$

where w_ℓ and $w_{n\ell}$ are the policyholder's final wealth in the loss and no-loss states. The no-insurance contract is

$$\bar{c} = (W - L, W). \quad (2)$$

- Full insurance means $w_\ell = w_{n\ell}$.
- Underinsurance means the policyholder's wealth is lower in the loss state than in the no-loss state.
- Contracts are exclusive in the baseline: the policyholder chooses at most one contract.
- Contracts are nonwithdrawable in the baseline: insurers cannot retract offers after seeing competitors' offers.

Interpretation: Exclusive contracts make the model directly relevant to ordinary insurance markets, where a policyholder typically cannot combine multiple mutually offsetting insurance contracts.

3.3 Policyholder utility

The policyholder is risk averse. Expected utility under loss probability p is

$$\mathbb{E}u_p(c) = pu(w_\ell) + (1 - p)u(w_{n\ell}). \quad (3)$$

For existence results, the paper specializes to CARA utility:

$$u(w) = \frac{1 - e^{-\beta w}}{\beta}, \quad (4)$$

where $\beta > 0$ is absolute risk aversion.

Interpretation: The qualitative characterization is more general, but CARA utility permits tractable threshold conditions for equilibrium existence.

3.4 Insurer profits

Insurers are risk neutral. If an insurer sells contract $c = (w_\ell, w_{n\ell})$ and believes the loss probability is p , expected profit is

$$\mathbb{E}\pi_p(c) = p(W - L - w_\ell) + (1 - p)(W - w_{n\ell}). \quad (5)$$

Interpretation: The insurer receives the premium implicitly embedded in the difference between the policyholder's initial wealth and final wealth, and pays the loss-state indemnity.

3.5 Signals and signal profiles

Each insurer receives a private signal $\hat{\theta}_i \in \{\hat{s}, \hat{d}\}$ about the environment. Signal precision is

$$\alpha = \Pr(\hat{\theta}_i = \hat{s} \mid \theta = s) = \Pr(\hat{\theta}_i = \hat{d} \mid \theta = d), \quad \alpha \in (1/2, 1). \quad (6)$$

The signal profile is summarized by

$$\sigma \in \{0, 1, 2\}, \quad (7)$$

where σ is the number of insurers who received the safe signal $\hat{s}$.

The policyholder's posterior loss probability after learning signal profile σ is

$$p_\sigma = \sum_{\theta \in \{d, s\}} p_\theta \Pr(\theta \mid \sigma). \quad (8)$$

- $\sigma = 0$: both insurers receive $\widehat{d}$.
- $\sigma = 1$: one insurer receives $\widehat{s}$ and one receives $\widehat{d}$.
- $\sigma = 2$: both insurers receive $\widehat{s}$.

Interpretation: The policyholder does not initially observe σ . Whether she learns it depends on whether insurers' contract menus reveal their private signals.

3.6 Timing

1. Nature chooses the environment θ and draws each insurer's signal.
2. Each insurer privately observes its own signal and updates its belief.
3. Insurers simultaneously offer public menus of contracts.
4. The policyholder observes all offers, forms beliefs, and chooses one contract or remains uninsured.
5. The accepted contract is implemented and payoffs are realized.

Interpretation: Insurers know only their own signal when they choose menus. The policyholder may infer the whole market signal profile from the menus, so she may become better informed than each individual insurer at the contract-choice stage.

4 Equilibrium concept and beliefs

4.1 Symmetric perfect Bayesian equilibrium

The paper focuses on symmetric pure-strategy perfect Bayesian equilibria. An equilibrium consists of:

- menus C_s^e and C_d^e chosen by safe-signal and dangerous-signal insurers;
- a policyholder choice rule;
- a belief mapping $\mu^e(C)$ after observing menus.

Sequential rationality requires:

1. each insurer's menu maximizes expected profits given its type and the other insurer's strategy;
2. the policyholder chooses the utility-maximizing contract given beliefs;
3. beliefs are consistent with Bayes' rule on the equilibrium path.

Interpretation: The key equilibrium object is not only the contract but also what the contract offer communicates.

4.2 Off-equilibrium beliefs and optimism

If an insurer deviates, the policyholder forms an off-path belief about the deviator's type. Let

$$\tilde{\mu} \in [0, 1] \tag{9}$$

be the probability that the deviating insurer is believed to be of type $\widehat{s}$.

- A larger $\tilde{\mu}$ means the policyholder is more optimistic about the deviating insurer's information.
- Full optimism corresponds to $\tilde{\mu} = 1$.
- Pessimism corresponds to $\tilde{\mu} = 0$.

Interpretation: Optimistic beliefs are powerful because they reduce the policyholder's willingness to pay for deviating contracts. If the deviator is likely safe-signal, the policyholder expects low risk and demands a low premium.

5 Noninformative equilibria

5.1 Definition and economic meaning

A noninformative equilibrium is one in which both insurer types offer the same menu:

$$C_{\hat{s}}^e = C_{\hat{d}}^e. \quad (10)$$

Because offers do not differ by signal, the policyholder learns nothing about insurers' signals. She therefore uses the ex ante loss probability

$$\bar{p} = p_d q + p_s (1 - q). \quad (11)$$

Interpretation: In noninformative equilibria, insurers pool on their types. The market outcome can look as if insurers have no private information, even though they do.

5.2 Equilibrium contract

Since there is only one perceived type of policyholder, the equilibrium menu degenerates to a single contract $c^e \in \Omega$.

The contract must satisfy three classes of constraints.

5.2.1 Insurer participation

For each insurer type $\hat{\theta} \in \{\hat{s}, \hat{d}\}$,

$$\mathbb{E}\pi_{p_{\hat{\theta}}}(c^e) \geq 0. \quad (12)$$

Interpretation: The contract must not lose money for either signal type. Since $\hat{d}$ insurers assess higher risk, their participation constraint is typically harder to satisfy.

5.2.2 Policyholder participation

The policyholder must prefer the contract to remaining uninsured:

$$\mathbb{E}u_{\bar{p}}(c^e) \geq \mathbb{E}u_{\bar{p}}(\bar{c}). \quad (13)$$

Interpretation: Because the policyholder learns nothing in equilibrium, this constraint uses the prior loss probability $\bar{p}$.

5.2.3 No-deviation condition

Let $c_{\hat{\theta}}^{dev}$ be the most profitable acceptable deviation for insurer type $\hat{\theta}$:

$$c_{\hat{\theta}}^{dev} = \arg \max_{c_i \in \Omega} \mathbb{E}\pi_{p_{\hat{\theta}}}(c_i) \quad \text{s.t.} \quad \mathbb{E}u_p(c_i) \geq \mathbb{E}u_p(\bar{c}), \quad (14)$$

where p is the loss probability induced by the policyholder's off-path belief.

The equilibrium must satisfy

$$\frac{1}{2} \mathbb{E}\pi_{p_{\hat{\theta}}}(c^e) \geq \mathbb{E}\pi_{p_{\hat{\theta}}}(c_{\hat{\theta}}^{dev}) \quad \text{for all } \hat{\theta} \in \{\hat{s}, \hat{d}\}. \quad (15)$$

Interpretation: The factor $1/2$ appears because in equilibrium each insurer sells the contract with equal probability. A deviation must not be more profitable than the expected equilibrium payoff.

5.3 Lemma 1: optimistic beliefs

Lemma 1. In any noninformative equilibrium, out-of-equilibrium beliefs are sufficiently optimistic:

$$\tilde{\mu} > 1 - q. \quad (16)$$

Read:

- If the policyholder sees a deviating offer, she must believe with sufficiently high probability that it comes from a safe-signal insurer.
- This makes her expect a lower loss probability and therefore demand a lower premium.
- Deviations that would otherwise undercut the equilibrium contract become unattractive or unacceptable.

Economic message: Optimistic beliefs weaken price competition. An insurer cannot easily attract the policyholder with a small undercut because the undercut itself signals low risk and reduces the policyholder's willingness to pay.

5.4 Proposition 1: existence of efficient noninformative equilibria

Under CARA utility, an ex ante efficient contract c is a noninformative equilibrium if and only if there exist thresholds β and $\bar{\beta}$ such that

$$\beta \leq \bar{\beta}, \quad \beta \in [\beta, \bar{\beta}]. \quad (17)$$

The set of noninformative equilibria E^* is nonempty if insurer signal precision α is large and the safe-environment loss probability p_s is small.

Read:

- A lower bound on risk aversion is needed so that the policyholder values insurance enough to support gains from trade.
- An upper bound on risk aversion is needed so deviations do not become too attractive.
- High signal precision and low safe-state risk make deviations less profitable when the policyholder is optimistic.

Economic message: Noninformative equilibria can be efficient because pooling can survive when deviations are interpreted optimistically and thus fail to attract demand.

5.5 Corollary 1: comparative statics of risk-aversion thresholds

If the policyholder's loss L is sufficiently large:

- the lower threshold β increases in signal precision α and in the probability q of the dangerous environment;
- the upper threshold $\bar{\beta}$ increases in α and decreases in q .

Interpretation: A more precise signal makes the dangerous signal more pessimistic and the safe signal more optimistic. This changes both participation and deviation incentives.

5.6 Economic properties of noninformative equilibria

- There can be multiple noninformative equilibrium contracts.
- They are always strictly profitable for safe-signal insurers.
- They can also be strictly profitable for dangerous-signal insurers.
- They may entail full insurance and ex ante Pareto efficiency.

Interpretation: This is the first major departure from standard competitive insurance intuition: profitable, efficient pooling can exist even with exclusive contracts.

6 Informative equilibria

6.1 Definition and signal revelation

In an informative equilibrium,

$$C_s^e \neq C_d^e. \quad (18)$$

The policyholder can infer the signal profile $\sigma \in \{0, 1, 2\}$ by observing the two menus.

- A safe-signal insurer can face signal profiles $\sigma \in \{1, 2\}$.
- A dangerous-signal insurer can face signal profiles $\sigma \in \{0, 1\}$.
- Therefore each insurer type offers a menu with one contract for each possible signal profile it may face.

Interpretation: Offers become signals. Once the policyholder observes both offers, she has market information about the signal profile that no individual insurer observed before offering its menu.

6.2 Screening created by signaling

Because the policyholder observes the signal profile through menus, the insurer's menu must screen policyholder beliefs across profiles. The policyholder's chosen contract must satisfy incentive compatibility:

$$c_{\theta, \sigma}^e = \arg \max_{c \in C_{\theta}^e} \mathbb{E} u_{p_{\sigma}}(c). \quad (19)$$

Insurers must also truthfully reveal their signals:

$$C_{\theta}^e = \arg \max_{C_i} \mathbb{E} \pi_{p_{\theta}}(C_i). \quad (20)$$

Interpretation: Informative equilibria combine signaling constraints for insurers and screening constraints for policyholders. This dual incentive problem is the source of inefficiency.

6.3 Proposition 2: characterization of informative equilibria

Proposition 2. Any informative equilibrium $C^e = (C_s^e, C_d^e) \in E^{**}$ has the following properties:

1. accepted contracts are fully separating across signal profiles;
2. contracts are incentive compatible for insurers and for the policyholder;
3. in $\sigma = 1$, the contract offered by the safe-signal insurer is accepted with probability one, entails full insurance, and yields strictly positive profits;
4. in $\sigma = 2$, accepted contracts entail underinsurance and are actuarially fair;
5. off-path beliefs are fully optimistic.

Read:

- In $\sigma = 2$, both insurers have safe signals. Competition drives profits to zero, but incentive compatibility requires underinsurance.
- In $\sigma = 1$, the safe-signal insurer has a competitive edge because the dangerous-signal insurer's menu is constrained by incentive compatibility with $\sigma = 0$.
- The safe-signal insurer's contract in $\sigma = 1$ can therefore be full insurance and strictly profitable.
- Full optimism is needed to rule out deviations that would otherwise be accepted under less favorable beliefs.

Economic message: Informative equilibria reveal information but generate inefficiency. The policyholder learns more, but the need to screen across signal profiles creates underinsurance.

6.4 Proposition 3: existence of informative equilibria

Under CARA utility, a specific informative equilibrium $\bar{C}^e$ exists if and only if there exist thresholds β and $\bar{\beta}$ such that

$$\beta \leq \bar{\beta}, \quad \beta \in [\beta, \bar{\beta}]. \quad (21)$$

The interval is nonempty if α and q are large.

Read:

- The lower bound on β comes from truthful-telling constraints.
- If risk aversion is too low, the safe-signal insurer earns too little in $\sigma = 1$ and may prefer to mimic a dangerous-signal insurer.
- The upper bound comes from cross-subsidy deviations.
- High α and high q make the relevant signal profiles sufficiently informative and help support equilibrium.

Economic message: Informative equilibria are harder to sustain than noninformative equilibria because they must satisfy both truth-telling and cross-subsidy constraints.

7 Role of signal precision

7.1 Why signal precision matters

Signal precision α changes insurers' posterior assessments of risk and changes the probability of signal profiles.

- A higher α makes a safe signal more reassuring.
- A higher α makes a dangerous signal more alarming.
- A higher α changes how likely mixed and homogeneous signal profiles are.

Interpretation: The effect of better information is not one-dimensional. It affects willingness to pay, profits, and the probability of profitable states.

7.2 Proposition 4: maximum premiums

Proposition 4.

1. In a maximum-premium noninformative equilibrium, increasing α does not affect the maximum equilibrium premium.
2. In a maximum-premium informative equilibrium, increasing α raises the maximum premium in signal profile 0, while leaving it unchanged in the other signal profiles.

Read:

- In a noninformative equilibrium, the policyholder's participation constraint uses the ex ante loss probability $\bar{p}$, which does not depend on α .
- In an informative equilibrium, signal profile 0 becomes more pessimistic as α rises, increasing the premium the policyholder is willing to pay.
- Signal profile 1 does not change in the same way, and signal profile 2 may be associated with autarky or actuarially fair outcomes.

Economic message: More precise information affects premiums only when information is revealed and the policyholder conditions on the signal profile.

7.3 Proposition 5: maximum profits

Proposition 5.

1. In noninformative equilibria, increasing α increases the maximum profit of safe-signal insurers and decreases the maximum profit of dangerous-signal insurers.
2. In informative equilibria, increasing α decreases the maximum profit of safe-signal insurers and increases the maximum profit of dangerous-signal insurers.

Read:

- In noninformative equilibria, premiums do not change with α , but safer signal types assess lower risk and therefore earn more.
- In informative equilibria, probability weights over signal profiles matter.
- Safe-signal insurers earn positive profits mainly in mixed signal profile $\sigma = 1$; as α rises, mixed profiles become less likely.
- Dangerous-signal insurers benefit because homogeneous dangerous profiles become more likely and more pessimistic.

Economic message: Better information can redistribute profits across insurer types in different directions depending on whether information is revealed.

7.4 Corollary 2: limiting case of perfect signals

As $\alpha \rightarrow 1$, informative equilibria still need not converge to full insurance in signal profile 2.

Read:

- Even nearly perfect information does not eliminate the screening distortion in informative equilibria.
- The contract in signal profile 2 remains constrained by incentive compatibility with the contract in signal profile 1.

Economic message: This result distinguishes heterogeneous information from the identical-information model of Villeneuve (2005). Perfectly precise but heterogeneous information can still generate inefficient informative equilibria.

8 Extensions

8.1 Withdrawable contracts

The baseline assumes that insurers cannot withdraw offers. The extension allows insurers to retract one or more contracts after observing competitors' offers.

Proposition 6. If contracts are withdrawable:

1. all baseline noninformative equilibria remain equilibria;
2. informative equilibria can exist in which outcomes are fully efficient in signal profiles 1 and 2, actuarially fair in signal profile 2, strictly profitable for safe-signal insurers, and possibly profitable for dangerous-signal insurers.

Read:

- In noninformative equilibria, observing the competitor's offer reveals no information, so withdrawal does not change the logic.
- In informative equilibria, observing offers reveals the signal profile, allowing insurers to withdraw loss-making contracts.
- Withdrawal improves efficiency because incentive constraints in some states become less binding.

Economic message: Contract withdrawal weakens the inefficiency of informative equilibria but does not eliminate the profitability or information-revelation logic.

8.2 Market concentration

The model extends from two insurers to $N > 2$ insurers.

For noninformative equilibria, the no-deviation condition becomes

$$\frac{1}{N} \mathbb{E} \pi_{p_{\hat{\theta}}}(c^e) \geq \mathbb{E} \pi_{p_{\hat{\theta}}}(c_{\hat{\theta}}^{dev}). \quad (22)$$

Proposition 7. There exists $\bar{N}$ such that for all $N > \bar{N}$, the set of noninformative equilibria is empty. For $N < \bar{N}$, the minimum expected profit among noninformative equilibrium contracts increases with N .

Proposition 8. There exists $\bar{\bar{N}}$ such that informative equilibria are empty for all $N > \bar{\bar{N}}$.

Read:

- With more insurers, each firm's equilibrium sale probability falls from $1/2$ to $1/N$.
- This makes equilibrium profits smaller relative to deviation profits.
- Equilibria may disappear if the market is too fragmented.

Economic message: The model provides a theoretical reason why insurance markets may remain concentrated: too many informed competitors can undermine equilibrium existence.

8.3 Policyholder private information

The Internet Appendix extends the model to allow the policyholder to have private information about her idiosyncratic risk, while insurers hold private information about the general environment.

- Noninformative equilibria can still entail risk pooling among policyholder types.
- Optimistic off-path beliefs can still block cream-skimming deviations.
- When insurers receive homogeneous signals, informative equilibria can resemble Rothschild-Stiglitz menus.

Interpretation: Two-sided asymmetric information does not automatically restore the classic no-pooling result. The signaling role of insurer deviations remains central.

9 Figures and original visual material

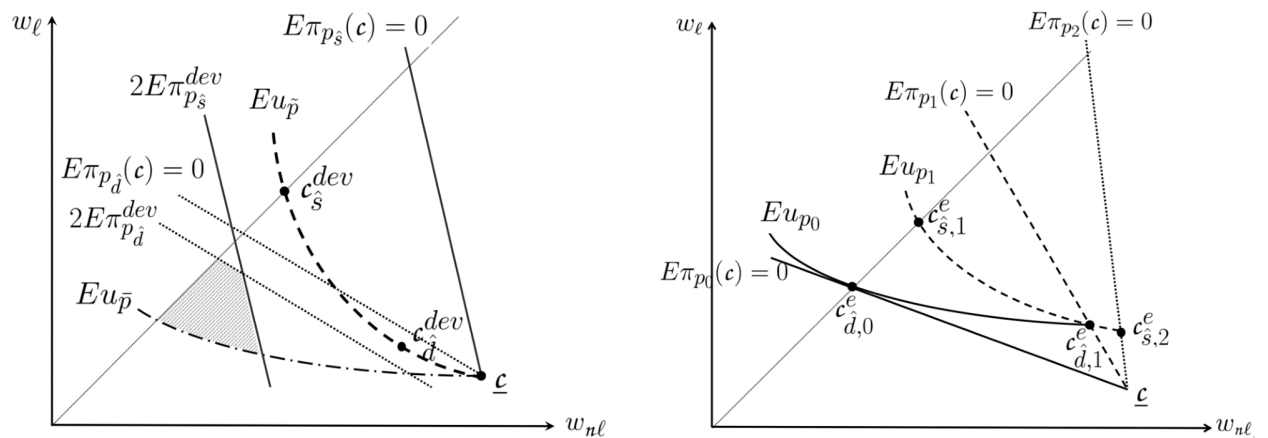
The paper is theoretical and contains two main figures. They are directly cropped from the paper and included below.

9.1 Figure 1: Set of noninformative equilibria

Read:

- The shaded region represents contracts that can be sustained as noninformative equilibrium outcomes.
- These contracts satisfy insurer participation, policyholder participation, and no-deviation constraints.
- The figure shows why multiple noninformative equilibrium contracts may exist.
- Some of these contracts can entail full insurance and ex ante efficiency.

Economic message: The figure illustrates the key nonstandard result: pooling can survive competition because deviations are interpreted through optimistic beliefs.



9.2 Figure 2: Informative equilibrium allocation

Read:

- The figure displays the menu structure in an informative equilibrium.
- A safe-signal insurer offers contracts for signal profiles 1 and 2.
- A dangerous-signal insurer offers contracts for signal profiles 0 and 1.
- The accepted contract in $\sigma = 1$ is the safe-signal insurer's full-insurance contract.
- The accepted contract in $\sigma = 2$ is actuarially fair but involves underinsurance.

Economic message: The figure makes clear why information revelation produces screening distortions. Once the policyholder learns the signal profile, menus must sort her inferred type across states.

10 Short summary

- The paper studies competition between insurers with private, imperfect, heterogeneous information about policyholder risk.
- Insurers may offer contracts that reveal or conceal their private signals.
- Noninformative equilibria pool insurer types and can be strictly profitable.
- Noninformative equilibria can also be efficient and provide full insurance.
- Informative equilibria reveal the signal profile but create an endogenous screening problem.
- Informative equilibria involve incentive compatibility constraints and underinsurance.
- Optimistic out-of-equilibrium beliefs are essential for sustaining profitable equilibria.
- More precise signals affect premiums and profits differently depending on whether information is revealed.
- Withdrawable contracts improve efficiency in informative equilibria but do not overturn noninformative equilibria.
- With too many insurers, equilibrium may fail to exist, providing a new rationale for market concentration.
- The paper's message is that big data can change the informational hierarchy in insurance markets, but competition need not force disclosure or zero profits.

11 Conclusion

11.1 Core conclusion

Abrardi, Colombo, and Tedeschi show that when insurers are privately and heterogeneously informed about policyholder risk, competitive insurance markets can support strictly profitable equilibria. Information may be revealed through menus, but it need not be. Pooling on insurer information can survive and can even be efficient.

11.2 Economic intuition

The economic intuition is that deviations themselves are informative. If the policyholder interprets a deviating contract as coming from a safe-signal insurer, she expects a lower loss probability and is unwilling to pay much for insurance. This makes undercutting deviations less attractive and weakens competitive pressure. When insurers reveal information, the policyholder learns the signal profile and the problem becomes a mixture of signaling and screening, generating underinsurance.

11.3 Takeaway

The paper suggests that data-rich insurance markets may not converge to full transparency or zero profits. Better insurer information can increase premiums and reshape profits, but whether information is revealed depends on equilibrium beliefs and contract design. In this sense, insurers may rationally benefit from pretending ignorance is bliss.